Sports Tournament Ranking System

Project Matter, Algorithm and Pseudocode

Assigned Syllabus Matter

Python solutions using only the assigned syllabus: computational thinking, algorithms, flowcharts, pseudocode, variables, data types, operators, conditional statements, loops, strings, pattern generation, lists, nested lists, tuples, tuple operations, and unpacking.

1. Aim

To create a tournament ranking system that accepts team match statistics, calculates points and goal difference for each team, and ranks all teams according to the competition rules.

2. Description

The program accepts the number of participating teams. For each team, it reads the team name, wins, draws, losses, goals scored, and goals conceded. It calculates points using 3 points for a win and 1 point for a draw. It calculates goal difference by subtracting goals conceded from goals scored. Each team's details are stored in a nested list. The list is then sorted in descending order using points, goal difference, and goals scored as tie-break criteria. Finally, the program displays the ranked teams.

3. Input

Input	Purpose
Number of teams	Specifies how many teams are included in the tournament.
Team name	Identifies each participating team.
Wins, draws and losses	Match results used to calculate tournament points.
Goals scored	Total goals scored by the team; also used as the third ranking criterion.
Goals conceded	Total goals conceded; used with goals scored to calculate goal difference.

4. Data Structure Used

The program uses a list named **team_data**. Each team is stored as an inner list containing four values in this order: team name, points, goal difference, and goals scored. This is a nested-list structure. During sorting, a tuple of the three numeric ranking values is used as the comparison key.

5. Competition and Ranking Rules

Points: Each win earns 3 points and each draw earns 1 point. Losses earn no points in this calculation.
Goal difference: Goals scored minus goals conceded.
Ranking priority: Teams are ordered by higher points first. If points are equal, higher goal difference ranks first. If both are equal, higher goals scored ranks first.

6. Algorithm

1. Start and input the number of teams.

2. Create an empty list named team_data.

3. Repeat for every team: input its name, wins, draws, losses, goals scored, and goals conceded.

4. Calculate points as 3 multiplied by wins plus draws.

5. Calculate goal difference as goals scored minus goals conceded.

6. Store the team name, points, goal difference, and goals scored in a team record; append it to team_data.

7. After all teams are entered, sort team_data in descending order by points, goal difference, and goals scored, in that priority.

8. Set rank to 1 and display the tournament rankings heading.

9. For each team in the sorted list, display its rank and statistics, then increase rank by 1.

10. Stop.

7. Pseudocode

```
START
    INPUT number_of_teams
```

```
    CREATE empty list team_data

    FOR each team i from 1 to number_of_teams
        INPUT team name
        INPUT wins, draws, losses
        INPUT goals_scored, goals_conceded

        CALCULATE points = (wins * 3) + draws
        CALCULATE goal_difference = goals_scored - goals_conceded

        CREATE team record containing:
            name, points, goal_difference, goals_scored
        APPEND team record to team_data
    END FOR

    SORT team_data in descending order using:
        1. points
        2. goal_difference
        3. goals_scored

    DISPLAY "Tournament Rankings"
    SET rank = 1

    FOR each team record in sorted team_data
        DISPLAY rank, team name, points,
                goal_difference, goals_scored
        INCREMENT rank by 1
    END FOR
STOP
```

8. Output

The program displays a ranked list of all teams. Each output line contains the team's rank, name, points, goal difference, and goals scored.

9. Important Notes

• The sorting order is descending for all three ranking values: points, goal difference, and goals scored.
• Losses are collected as input but are not directly used in the points formula or ranking key.
• If two or more teams have identical points, goal difference, and goals scored, the supplied code does not define an additional tie-break rule. Python's sort is stable, so teams with identical keys retain their original input order.
• The program assumes that the entered number of teams and match statistics are valid non-negative integer values.

10. Conclusion

The Sports Tournament Ranking System demonstrates how lists, nested lists, loops, arithmetic operators, and sorting can be used to organize competition results. It applies a clear set of ranking rules to produce an ordered tournament table.